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CONTROL DEVICE, MACHINE TOOL, AND CONTROL METHOD

Abstract

A control device of a machine tool comprising a table that supports an object to be machined, a main shaft on which a tool is mounted, and a mist collector that collects mist comprises: an acquisition unit that acquires at least one piece of information from among a machining load, a relative movement speed of the main shaft in relation to the table, and a rotational speed of the main shaft; a startup determination unit that determines whether to start the mist collector on the basis of the at least one piece of acquired information; and a collector control unit that automatically controls the mist collector on the basis of the determination details.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a control device, a machine tool, and a control method.

BACKGROUND ART

[0002] A mist collector collects mist generated within a machining area of a machine tool (refer also to JP 2012-076006 A). The mist collector collects the mist within the machining area, and prevents the mist from leaking out to the exterior of the machining area. The mist is a coolant of fine or minute particles that are suspended within the air.

SUMMARY OF THE INVENTION

[0003] The mist collector vibrates when operated. Depending on the content of machining performed by the machine tool, vibration of the mist collector may have a significant adverse effect on the machine accuracy of the machine tool.

[0004] An object of the present invention is to solve the aforementioned problem.

[0005] A first aspect of the present invention is a control device for a machine tool including a table supporting a workpiece to be machined by a tool in a machining area, a spindle to which the tool is mounted and which rotates and moves relative to the table, and a mist collector configured to collect mist in the machining area, the control device including an acquisition unit configured to acquire at least one type of information from among a machining load applied to the tool by performing machining on the workpiece, a relative movement speed of the spindle with respect to the table, and a rotational speed of the spindle, a start determination unit configured to determine whether or not to start the mist collector based on the at least one type of information acquired by the acquisition unit, and a collector control unit configured to automatically control the mist collector based on a determination content of the start determination unit.

[0006] A second aspect of the present invention is a machine tool including the control device according to the first aspect.

[0007] A third aspect of the present invention is a control method for a machine tool including a table supporting a workpiece to be machined by a tool in a machining area, a spindle to which the tool is mounted and which rotates and moves relative to the table, and a mist collector configured to collect mist in the machining area, the control method including an acquisition step in which a computer acquires at least one type of information from among a machining load applied to the tool by performing machining on the workpiece, a relative movement speed of the spindle with respect to the table, and a rotational speed of the spindle, a start determination step in which the computer determines whether or not to start the mist collector based on the at least one type of information acquired in the acquisition step, and a collector control step in which the computer controls the mist collector based on a determination content in the start determination step.

[0008] According to the present invention, since the mist collector is automatically controlled in accordance with the content of processing, vibration of the mist collector is prevented from having a large adverse effect on the machine accuracy.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a schematic diagram of a machine tool according to an embodiment;

[0010] FIG. 2 is a block diagram of a control device;

[0011] FIG. 3 is a flowchart illustrating a control method according to an embodiment;

[0012] FIG. 4 is a schematic diagram of a machine tool according to an exemplary modification 1;

[0013] FIG. 5 is a block diagram of a control device according to an exemplary modification 2; and

[0014] FIG. 6 is a block diagram of a control device according to an exemplary modification 3.
DETAILED DESCRIPTION OF THE INVENTION

Embodiment

[0015] FIG. 1 is a schematic diagram of a machine tool 10 according to an embodiment.

[0016] The X direction and the Y direction shown in FIG. 1 are directions that are parallel to a horizontal plane. The X direction and the Y direction are mutually orthogonal with each other. The Z direction shown in FIG. 1 is a direction that is parallel to the direction of gravity. Accordingly, the Z direction is perpendicular to the X direction and the Y direction. However, the Z direction shown in FIG. 1 indicates a direction that is opposite to the direction of gravity.

[0017] The machine tool 10 includes a machining device 12 and a control device 14.

[0018] The machining device 12 is a machine that carries out machining on a workpiece using a tool 16. The machining device 12 is equipped with a spindle 18, a spindle head 20, a column 22, a pedestal 24, a table 26, a table drive unit 28, a cover 30, a coolant supply device 32, and a mist collector 34.

[0019] A tool holder 36 is attached to the spindle 18 (refer to FIG. 1). The tool holder 36 is capable of being attached and detached to and from the spindle 18. The tool holder 36 retains the tool 16. The tool 16, for example, is a spring-necked turning tool, a drill, an end mill, a milling cutter, or the like.

[0020] The machining device 12 is further equipped with a tool magazine 38. The tool magazine 38 detachably retains a plurality of the tools 16. One of the plurality of the tools 16 that are retained in the tool magazine 38 is changeably attached to the tool holder 36.

[0021] The spindle head 20 supports the spindle 18. The spindle head 20 is provided with a spindle motor 21 for rotating the spindle 18. The spindle motor 21 is, for example, a spindle motor. The spindle motor 21 includes a shaft (not shown). The tool 16 mounted on the spindle 18 via the tool holder 36 rotates in accordance with the rotation of the shaft of the spindle motor 21.

[0022] The spindle motor 21 includes a torque sensor 23 and an encoder 25. The torque sensor 23 outputs a detection signal corresponding to the output torque of the spindle motor 21. The encoder 25 is a rotary encoder. The encoder 25 outputs a detection signal corresponding to the rotational position of the shaft of the spindle motor 21. The detection signal of the torque sensor 23 and the detection signal of the encoder 25 are input to the control device 14. The control device 14 will be described in more detail later.

[0023] The column 22 supports the spindle head 20. Further, the column 22 includes a motor that causes the spindle head 20 to be moved in the Z direction. The column 22 is supported on the pedestal 24.

[0024] The pedestal 24 is installed on an installation surface. The installation surface, for example, is a floor of a factory. The installation surface may be a support surface of a platform that is provided on the floor. The installation surface extends, for example, parallel to the horizontal plane. The pedestal 24 may be equipped with a plurality of leg members 24a. Each of the leg members 24a, for example, may be a caster, a jack, or the like.

[0025] The table drive unit 28 is supported on the pedestal 24. The table drive unit 28 includes a first slide unit 42, a saddle 44, a second slide unit 46, and a plurality of feed axis motors 47 (47X, 47Y). The plurality of feed axis motors 47 include a Y-axis motor 47Y and an X-axis motor 47X.

[0026] The first slide unit 42 is provided on the pedestal 24. The first slide unit 42 includes, for example, guide rails that extend in the Y direction. The first slide unit 42 supports the saddle 44.

[0027] The saddle 44 moves in the Y direction in response to the driving of the Y-axis motor 47Y. The Y-axis motor 47Y is controlled by the control device 14. The saddle 44 moves while being guided by the first slide unit 42.

[0028] The Y-axis motor 47Y is, for example, a servo motor. The Y-axis motor 47Y includes a shaft 49Y and an encoder 29 (29Y). The shaft 49Y rotates in accordance with a drive current supplied to the Y-axis motor 47Y. The encoder 29Y is a rotary encoder. The encoder 29Y outputs a detection

signal corresponding to the rotational position of the shaft **49Y**. The detection signal of the encoder **29Y** is input to the control device **14**.

[0029] The second slide unit **46** is provided on the saddle **44**. The second slide unit **46** includes, for example, guide rails that extend in the X direction.

[0030] The table **26** supports a non-illustrated workpiece downwardly of the spindle **18**. The table **26** is supported on the second slide unit **46**. The table **26** moves in the X direction in response to the driving of the X-axis motor **47X**. The X-axis motor **47X** is controlled by the control device **14**. The table **26** moves while being guided by the second slide unit **46**.

[0031] The X-axis motor **47X** is, for example, a servo motor. The X-axis motor **47X** includes a shaft **49X** and an encoder **29** (**29X**). The shaft **49X** rotates in accordance with a drive current supplied to the X-axis motor **47X**. The encoder **29X** is a rotary encoder. The encoder **29X** outputs a detection signal corresponding to the rotational position of the shaft **49X**. The detection signal of the encoder **29X** is input to the control device **14**.

[0032] The cover **30** covers the spindle **18**, the spindle head **20**, the column **22**, the pedestal **24**, the table **26**, and the table drive unit **28**. Consequently, the cover **30** forms a machining area **48**. A workpiece is machined within the machining area **48**.

[0033] The cover **30** is further equipped with a non-illustrated door and a non-illustrated window. The operator can carry out an introduction operation of the workpiece into the machining area **48** through the door that is in an opened state. Further, the operator can easily confirm the condition within the machining area **48** via the window.

[0034] The coolant supply device **32** is a device that supplies the coolant to the machining area **48**. The coolant supply device **32** is equipped with a coolant tank **50**, a nozzle **52**, a supply pipe **54**, and a pump **56**.

[0035] The coolant tank **50** stores the coolant. The coolant tank **50** is provided externally of the machining area **48**.

[0036] The nozzle **52** is a discharge unit that discharges the coolant. The nozzle **52** is disposed within the machining area **48**. Moreover, the coolant supply device **32** may be equipped with a plurality of the nozzles **52**.

[0037] The supply pipe **54** is a pipe that connects the coolant tank **50** and the nozzle **52**. Moreover, the coolant supply device **32** may be equipped with a plurality of the supply pipes **54**. The number of the supply pipes **54** is determined, for example, in accordance with the number of the nozzles **52**. The supply pipe **54** passes through the cover **30**, and connects the coolant tank **50** and the nozzle **52**.

[0038] The pump **56** is connected to the supply pipe **54**. The pump **56** draws in the coolant within the coolant tank **50**, and delivers the coolant to the nozzle **52**. Consequently, the coolant is discharged from the nozzle **52** into the machining area **48**. Moreover, the pump **56** is controlled by the control device **14**.

[0039] The coolant that is discharged into the machining area **48** cools the tool **16** and the workpiece. When machining is carried out in the machining area **48**, a mist of the coolant is generated. There is a concern that the mist may leak out to the exterior of the machining area **48** via small gaps that occur in the machining device **12**.

[0040] The mist collector **34** is a device that serves to collect the mist within the machining area **48**. The mist collector **34** is provided outside the machining area **48**. Further, the mist collector **34** is connected to the cover **30** via a duct **58**. The mist collector **34** collects the mist by drawing in air within the machining area **48**. In accordance with this feature, the mist is prevented from leaking out to the exterior of the machining area **48**.

[0041] By the tool **16** cutting the workpiece, fine or minute cutting chips are generated in the form of powdery dust within the machining area **48**. There is a concern that the powdery dust, in the same manner as the mist, may leak out to the exterior of the machining area **48** via small gaps that occur in the machining device **12**. The mist collector **34**, by drawing in air within the machining

area **48**, may collect not only the mist, but also the powdery dust. In accordance with this feature, the powdery dust is prevented from leaking out to the exterior of the machining area **48**.

[0042] The mist collector **34** may be connected to the coolant tank **50**. Consequently, the mist that is collected by the mist collector **34** can be returned, as the coolant, to the coolant tank **50**.

[0043] In the case that the mist collector **34** and the coolant tank **50** are connected to each other, it is preferable for the mist collector **34** and the coolant tank **50** to be connected to each other via a non-illustrated filtering device (filter). The filtering device removes impurities in the coolant sent from the mist collector **34** to the coolant tank **50**. By connecting the mist collector **34** and the coolant tank **50** via the filtering device, a clean coolant can be returned from the mist collector **34** to the coolant tank **50**. Impurities in the coolant, for example, are the cutting chips that have been collected together with the mist.

[0044] FIG. **2** is a block diagram of the control device **14**.

[0045] The control device **14** is a computer that controls the machining device **12**. The control device **14**, for example, is a numerical control device. The control device **14** is equipped with a display unit **60**, an operation unit **62**, a storage unit **64**, a computation unit **66**, and a standby electrical power supply unit **68**.

[0046] The display unit **60** is a display device equipped with a display screen **60d**. The display unit **60**, for example, is a liquid crystal display device or an OEL (Organic Electro-Luminescence) display device.

[0047] The operation unit **62** is an input device that receives instructions from the operator to the control device **14**. The operation unit **62** includes, for example, an operation panel **62a**, and a touch panel **62b** or the like. The touch panel **62b** is provided on the display screen **60d**. The operation unit **62** (the operation panel **62a**) may be equipped with a keyboard, a mouse, or the like.

[0048] The storage unit **64** is constituted by a non-illustrated volatile memory, and a non-illustrated nonvolatile memory. As an example of the volatile memory, there may be cited a random access memory (RAM) or the like. As an example of the nonvolatile memory, there may be cited a ROM (Read Only Memory) and a flash memory or the like. Data and the like may be stored, for example, in the volatile memory. A program, a data table, a map and the like may be stored, for example, in the nonvolatile memory. At least a portion of the storage unit **64** may be provided in the aforementioned processor, an integrated circuit, or the like. The storage unit **64** stores a control program **70**, a machining program **72**, and a plurality of threshold values TH (TH1, TH2, TH3).

[0049] The control program **70** is a program for the purpose of causing the control device **14** to execute the control method according to the present embodiment. The control method will be described in more detail later.

[0050] The machining program **72** is a program including control commands for the machining device **12**. The machining program **72** includes, for example, a plurality of control commands for controlling the plurality of motors (**21**, **47**). Further, the machining program **72** also includes, for example, a plurality of control commands for the purpose of controlling the coolant supply device **32**. The machining program **72** is created or edited in advance by the operator.

[0051] The plurality of threshold values TH include a threshold value TH1 for a machining load LO, a threshold value TH2 for a relative movement speed VR, and a threshold value TH3 for the rotational speed VS of the spindle **18**. The machining load LO is a load applied to the tool **16** for machining the workpiece. The relative movement speed VR is a relative movement speed of the spindle **18** with respect to the table **26**. The relative movement speed VR includes a relative movement speed VRX and a relative movement speed VRY. The relative movement speed VRX is the relative movement speed of the spindle **18** in the X direction with respect to the table **26**. The relative movement speed VRY is a relative movement speed of the spindle **18** in the Y direction with respect to the table **26**.

[0052] The threshold value TH1 is, for example, the maximum allowable value of the machining load LO in predetermined machining. The predetermined machining refers to machining in which

the vibration of the mist collector **34** greatly affects the machine accuracy. The predetermined machining includes, for example, finishing, precision machining, or the like. The threshold value TH2 is, for example, a maximum allowable value of the relative movement speed VR in the predetermined machining. The storage unit **64** may store the threshold value TH2 for the relative movement speed VRX and the threshold value TH2 for the relative movement speed VRY. In this case, the threshold value TH2 for the relative movement speed VRX and the threshold value TH2 for the relative movement speed VRY may be equal to or different from each other. The threshold value TH3 is, for example, a maximum allowable value of the rotational speed VS of the spindle **18** in the predetermined machining. The specific values of the respective threshold values TH (TH1 to TH3) are determined based on experiments. The specific values of the respective threshold values TH (TH1 to TH3) may be provided to the operator from the manufacturer of the machine tool **10**.

[0053] The computation unit **66** is constituted by a processor including, for example, a CPU (Central Processing Unit), and a GPU (Graphics Processing Unit) or the like. More specifically, the computation unit **66** can be constituted by a processing circuit (Processing Circuitry).

[0054] The computation unit **66** includes an acquisition unit **73**, a machining control unit **74**, a start determination unit **75**, and a collector control unit **76**. The acquisition unit **73**, the machining control unit **74**, the start determination unit **75**, and the collector control unit **76** are implemented by the computation unit **66** executing the control program **70**. At least a portion of the acquisition unit **73**, the machining control unit **74**, the start determination unit **75**, and the collector control unit **76** may be realized by an integrated circuit such as an ASIC (Application Specific Integrated Circuit) or an FPGA (Field-Programmable Gate Array). At least a portion of the acquisition unit **73**, the machining control unit **74**, the start determination unit **75**, and the collector control unit **76** may be configured by an electronic circuit including a discrete device.

[0055] The acquisition unit **73** acquires at least one type of information from among the machining load LO, the relative movement speed VR, and the rotational speed VS. When acquiring the relative movement speed VR, the acquisition unit **73** may acquire both the relative movement speed VRX and the relative movement speed VRY, or may acquire only one of them. When acquiring only one of the relative movement speed VRX and the relative movement speed VRY, the acquisition unit **73** may acquire the one having the larger absolute value.

[0056] The machining load LO is calculated based on the torque of the spindle motor **21**. The torque of the spindle motor **21** is calculated based on a detection signal output from the torque sensor **23**. Therefore, the acquisition unit **73** can acquire the information of the machining load LO based on the detection signal output from the torque sensor **23**.

[0057] The relative movement speed VRX is calculated based on the rotational speed of the shaft **49X**. The rotational speed of the shaft **49X** is calculated based on a detection signal output from the encoder **29X**. Therefore, the acquisition unit **73** can acquire the information of the relative movement speed VRX based on the detection signal output from the encoder **29X**.

[0058] The relative movement speed VRY is calculated based on the rotational speed of the shaft **49Y**. The rotational speed of the shaft **49Y** is calculated based on a detection signal output from the encoder **29Y**. Accordingly, the acquisition unit **73** can acquire the information of the relative movement speed VRY based on the detection signal output from the encoder **29Y** to the control device **14**.

[0059] The rotational speed VS is calculated based on the rotational speed of the shaft of the spindle motor **21**. The rotational speed of the shaft of the spindle motor **21** is calculated based on a detection signal output from the encoder **25**. Therefore, the acquisition unit **73** can acquire the information of the rotational speed VS based on the detection signal output from the encoder **25**.

[0060] The machining control unit **74**, by controlling the machining device **12** based on the machining program **72**, carries out machining on the workpiece. For example, the machining control unit **74** controls the spindle motor **21**, the plurality of feed axis motors **47**, and the like based on the machining program **72**. However, within the machining device **12**, the mist collector

34 is controlled by the collector control unit **76**.

[0061] The start determination unit **75** determines whether or not a predetermined start condition is satisfied based on the information (the machining load **LO**, the relative movement speed **VR**, and the rotational speed **VS**) acquired by the acquisition unit **73**. The predetermined start condition is satisfied when at least one of the machining load **LO**, the relative movement speed **VR**, and the rotational speed **VS** acquired by the acquisition unit **73** is equal to or greater than the threshold value **TH**.

[0062] The start determination unit **75** determines whether at least one of (1) to (3) described below is satisfied in order to determine whether a predetermined start condition is satisfied: (1) Whether or not the machining load **LO** is equal to or greater than the threshold value **TH1**; (2) Whether or not the relative movement speed **VR** is equal to or greater than the threshold value **TH2**; and (3) Whether or not the rotational speed **VS** is equal to or greater than the threshold value **TH3**.

[0063] In the case that at least one of the above conditions (1) to (3) is satisfied, the start determination unit **75** determines that the predetermined start condition is satisfied. When it cannot be determined that the predetermined start condition is satisfied, the start determination unit **75** determines that the predetermined start condition is not satisfied.

[0064] When the predetermined start condition is satisfied, the start determination unit **75** determines to start the mist collector **34**. When the predetermined start condition is not satisfied, the start determination unit **75** determines not to start the mist collector **34**.

[0065] The collector control unit **76** controls the mist collector **34** based on a determination content of the start determination unit **75**.

[0066] For example, when the start determination unit **75** determines that the mist collector **34** is to be started, the collector control unit **76** automatically starts the mist collector **34** and causes the mist collector **34** to collect the mist in the machining area **48**.

[0067] On the other hand, when the start determination unit **75** determines that the mist collector **34** is not to be started, the collector control unit **76** does not automatically start the mist collector **34**. If the start determination unit **75** determines that the mist collector **34** is not to be started and the mist collector **34** is operating, the collector control unit **76** causes the mist collector **34** to be stopped.

[0068] According to the collector control unit **76**, the mist collector **34** is automatically started when a predetermined start condition is satisfied. As a result, the mist, powdery dust, etc. in the machining area **48** are collected by the mist collector **34**.

[0069] The mist collector **34** is vibrated when operated. The vibration of the mist collector **34** may cause vibration of the workpiece. However, when the predetermined start condition is satisfied, there is a high possibility that the machining is performed in which the vibration of the mist collector **34** does not greatly affect the machine accuracy. Therefore, when the predetermined start condition is satisfied, the vibration of the mist collector **34** is unlikely to adversely affect the machine accuracy.

[0070] The machining in which the vibration of the mist collector **34** does not greatly affect the machine accuracy is, for example, the machining in which the vibration stronger than the vibration generated by the mist collector **34** is generated from the portion of the machining device **12** other than the mist collector **34**. The machining may be, for example, rough machining. The rough machining tends to require a larger amount of cutting than the finishing. Accordingly, the rough machining tends to generate a large amount of mist, powdery dust, and the like in the machining area **48**. That is, in the case that a large amount of mist, powdery dust, or the like is generated in the machining area **48**, the control device **14** can automatically start the mist collector **34**.

[0071] In the case that the predetermined start condition is not satisfied, there is a high possibility that the vibration of the mist collector **34** is causing the machining to have a large influence on the machine accuracy. In this regard, according to the collector control unit **76**, in the case that the predetermined start condition is not satisfied, the mist collector **34** does not operate. This prevents the mist collector **34** from vibrating the workpiece in the case that a predetermined start condition

is not satisfied. Therefore, the possibility of a significant deterioration in the machine accuracy is reduced. As described above, the vibration of the mist collector **34** greatly affects the machine accuracy in, for example, finishing, precision machining, and the like. In the finishing, the precision machining, and the like, there is a tendency that the mist, the powdery dust, and the like generated in the machining area **48** are small. That is, in the case that the amount of mist, powdery dust, or the like generated in the machining area **48** is small, the control device **14** can automatically stop the mist collector **34**. In accordance with this feature, the power consumption of the mist collector **34** is suppressed.

[0072] It is preferable for the acquisition unit **73** to sequentially acquire at least one of the machining load LO, the relative movement speed VR, and the rotational speed VS while the machining control unit **74** performs machining. While the machining control unit **74** is performing machining, the start determination unit **75** preferably sequentially determines whether or not to start the mist collector **34** by using the latest information acquired by the acquisition unit **73**. Thus, the collector control unit **76** can appropriately turn ON and OFF the mist collector **34** in accordance with a change in at least one of the machining load LO, the relative movement speed VR, and the rotational speed VS during machining.

[0073] While the machining control unit **74** is performing machining, the acquisition unit **73** preferably acquires all the information of the machining load LO, the relative movement speed VR, and the rotational speed VS. By using all the information of the machining load LO, the relative movement speed VR, and the rotational speed VS, the start determination unit **75** can accurately determine whether or not the predetermined start condition is satisfied.

[0074] The standby electrical power supply unit **68** is an electrical power supply that is separate from the main power supply of the control device **14**. The standby electrical power supply unit **68** includes, for example, a battery. The standby electrical power supply unit **68** is integrated in the control device **14**. However, the standby electrical power supply unit **68** may be provided in the machine tool **10** as an external power supply for the control device **14**. Moreover, it should be noted that the main power supply for the control device **14** is not shown in the drawings.

[0075] In the case that the main power supply of the control device **14** is turned OFF while the mist collector **34** is in operation, the standby electrical power supply unit **68** supplies electrical power to each of the components of the control device **14**. In accordance with this feature, even after the main power supply has been turned OFF, the collector control unit **76** can continue to control the mist collector **34**.

[0076] For example, the main power supply of the control device **14** may be turned OFF before the collector control unit **76** causes the mist collector **34** to be stopped. In such a case, by supplying electrical power from the standby electrical power supply unit **68**, the collector control unit **76** can cause the mist collector **34** to be automatically stopped even after the main power supply of the control device **14** is turned OFF. In accordance with this feature, the wasteful power consumption of the mist collector **34** is suppressed.

[0077] FIG. **3** is a flowchart illustrating the control method according to the embodiment.

[0078] The control device **14** can execute a control method, for example, as illustrated in FIG. **3**. The control device **14** executes the control method of FIG. **3** while the machining control unit **74** performs machining based on the machining program **72**. Moreover, at the starting time of the control method, the mist collector **34** is stopped. The control method of FIG. **3** includes an acquisition step **S1**, a start determination step **S2**, a collector control step **S3**, and an end determination step **S4**. The collector control step **S3** includes a collector starting step **S31** and a collector stopping step **S32**.

[0079] In the acquisition step **S1**, the acquisition unit **73** acquires at least one type of information from among the machining load LO, the relative movement speed VR, and the rotational speed VS. It is preferable for the acquisition unit **73** to acquire all the information of the machining load LO, the relative movement speed VR, and the rotational speed VS.

[0080] In the start determination step S2, the start determination unit 75 determines whether or not to start the mist collector 34. In order to determine whether or not to start the mist collector 34, the start determination unit 75 determines whether or not a predetermined start condition is satisfied based on the information (LO, VR, VS) acquired by the acquisition unit 73 in the acquisition step S1.

[0081] When the predetermined start condition is satisfied (S2: YES), the start determination unit 75 determines to start the mist collector 34. When the predetermined start condition is not satisfied (S2: NO), the start determination unit 75 determines not to start the mist collector 34. The collector starting step S31 or the collector stopping step S32 is started in accordance with the determination content in the start determination step S2.

[0082] In the case that it is determined in the start determination step S2 that the mist collector 34 is to be started, the collector starting step S31 is started. In the collector starting step S31, the collector control unit 76 controls the mist collector 34 to automatically start the mist collector 34. If the mist collector 34 is already operating, the collector control unit 76 maintains the operating state of the mist collector 34.

[0083] When it is determined in the start determination step S2 that the mist collector 34 is not started, the collector stopping step S32 is started. In the collector stopping step S32, the collector control unit 76 controls the mist collector 34 to cause the mist collector 34 to be automatically stopped. In the case that the mist collector 34 has already been stopped, the collector control unit 76 maintains the stopped state of the mist collector 34.

[0084] In the end determination step S4, the machining control unit 74 determines whether or not the machining based on the machining program 72 is completed. If the machining is not finished (S4: NO), the flow from the acquisition step S1 to the end determination step S4 is repeated. When the machining is finished (S4: YES), the control method of FIG. 3 is brought to an end. In the case that the machining is finished and the mist collector 34 is operating, the collector control unit 76 causes the mist collector 34 to be automatically stopped.

Exemplary Modifications

[0085] Hereinafter, a description will be given concerning exemplary modifications of the above-described embodiment. However, any descriptions that are duplicative or overlap with those of the above-described embodiment will be appropriately omitted in the following description. Unless otherwise specified, the same reference numerals as in the above-described embodiment are used in referring to the elements that have already been described in the embodiment.

Exemplary Modification 1

[0086] FIG. 4 is a schematic diagram of a machine tool 101 (10) according to a first exemplary modification.

[0087] The machine tool 101 is further equipped with a sub-control device 78. In the machine tool 101, the standby electrical power supply unit 68 of the control device 14 may be omitted.

[0088] The sub-control device 78 is a computer that is separate from the control device 14. The sub-control device 78, for example, is equipped with a processor and a memory. The sub-control device 78 may also be equipped with an integrated circuit, a discrete device, or the like.

[0089] In the case that the control device 14 is stopped, the sub-control device 78 controls the mist collector 34 instead of the collector control unit 76. Therefore, even if the control device 14 is stopped, the mist collector 34 is controlled by the sub-control device 78 in the same manner as in the embodiment.

[0090] For example, if the main power supply of the control device 14 is turned OFF before the collector control unit 76 causes the mist collector 34 to be stopped, the sub-control device 78 can cause the mist collector 34 to be stopped, instead of the control device 14.

[0091] Moreover, it is preferable for the sub-control device 78 and the control device 14 to communicate with each other as appropriate, and to share the data necessary for controlling the mist collector 34. For example, the sub-control device 78 and the control device 14 share the

information acquired by the acquisition unit **73**, the determination content by the start determination unit **75**, or the progress of the machining. In accordance with these features, the sub-control device **78** is capable of smoothly taking over the control that was being performed by the collector control unit **76**. According to the present exemplary modification, even after the control device **14** is stopped, the control of the mist collector **34** can be continued by the sub-control device **78**.

Exemplary Modification 2

[0092] FIG. **5** is a block diagram of a control device **142** (**14**) according to an exemplary modification **2**.

[0093] The control device **142** is further equipped with an alarm output unit **80**.

[0094] The alarm output unit **80** outputs an alarm in the case that an abnormality has occurred in the machine tool **10**. For example, the machine tool **10** is appropriately provided with non-illustrated sensors for the purpose of detecting malfunctions or troubles in each of the respective components such as the spindle **18**, the spindle head **20**, the table drive unit **28**, etc. The alarm output unit **80** determines whether or not a malfunction has occurred in the machine tool **10** based on the signals output by the sensors. In the case that a malfunction is detected in any of the respective components of the machine tool **10**, the alarm output unit **80** issues a notification to the operator, for example via the display unit **60**, to the effect that a malfunction has occurred.

[0095] In the case that the alarm output unit **80** has output an alarm prior to the start of the machining, the machining control unit **74** does not start the machining until the cause of the alarm is eliminated. Further, in the case that the alarm output unit **80** has output an alarm after the start of the machining, the machining control unit **74** suspends the machining based on the machining program **72** until the cause of the alarm is eliminated.

[0096] In the case that the alarm output unit **80** outputs an alarm, the collector control unit **76** prohibits the operation of the mist collector **34** until the cause of the alarm is eliminated, regardless of the determination content of the start determination unit **75**. In the case that the mist collector **34** is operating at the time when the alarm is output, the collector control unit **76** causes the mist collector **34** to be stopped regardless of the determination content of the start determination unit **75**.

[0097] According to the present exemplary modification, in the case that an abnormality has occurred in the machine tool **10**, the mist collector **34** is prevented from operating.

Exemplary Modification 3

[0098] FIG. **6** is a block diagram of a control device **143** (**14**) according to an exemplary modification **3**.

[0099] The control device **143** further includes an estimation unit **82**.

[0100] The estimation unit **82** estimates (calculates) at least one of the estimated values of the machining load LO, the relative movement speed VR, and the rotational speed VS based on the machining program **72**.

[0101] For example, the machining program **72** includes a control command to drive the spindle motor **21** at a predetermined torque. The estimation unit **82** estimates the machining load LO based on the control command. Further, for example, a control command to rotate the shaft **49X** at a predetermined speed is included in the machining program **72**. The estimation unit **82** estimates the relative movement speed VRX based on the control command. Further, for example, a control command to rotate the shaft of the spindle motor **21** at a predetermined speed is included in the machining program **72**. The estimation unit **82** estimates the rotational speed VS based on the control command.

[0102] The estimation unit **82** estimates at least one of the estimated values of the machining load LO, the relative movement speed VR, and the rotational speed VS by using a mathematical equation. The mathematical equation is predetermined based on experiments. The estimation unit **82** may use different mathematical equations for the estimation of the machining load LO, the estimation of the relative movement speed VR, and the estimation of the rotational speed VS.

[0103] The estimation unit **82** may refer to information other than the machining program **72** as appropriate in order to estimate at least one of the machining load LO, the relative movement speed VR, and the rotational speed VS. For example, the machining load LO varies depending on the material of a workpiece or the tool **16**. Therefore, the estimation unit **82** may refer to the material of the workpiece or the tool **16** in order to calculate the estimated value of the machining load LO. In this case, the storage unit **64** may store the material of the workpiece or the tool **16**.

[0104] The acquisition unit **73** may acquire the value estimated by the estimation unit **82** as information.

[0105] According to the present modification, the start determination unit **75** can determine whether or not to start the mist collector **34** before the machining is started. That is, the estimation unit **82** can estimate (calculate) at least one of the estimated values of the machining load LO, the relative movement speed VR, and the rotational speed VS based on the machining program **72**, before machining is started. The acquisition unit **73** can acquire the estimated value before the machining is started. Thus, the start determination unit **75** can determine whether to start the mist collector **34** based on the comparison between the estimated value and the threshold value TH, before the machining is started.

Combination of Plural Exemplary Modifications

[0106] The above-described plurality of exemplary modifications may be combined as appropriate within a range in which there are no contradictions therebetween.

Modified Embodiments

[0107] It should be noted that the present invention is not limited to the embodiment described above, and various alternative or additional configurations may be adopted therein without departing from the essence and gist of the present invention.

[0108] For example, according to the above embodiment, the mist collector **34** stops when the machining control unit **74** completes the machining. However, the collector control unit **76** may control the mist collector **34** to cause the mist collector **34** to collect the mist in the machining area **48** until a predetermined time period has elapsed from the completion of the machining. By intentionally operating the mist collector **34** after the completion of the machining, it is possible to prevent a loss in the collection of the mist. The predetermined time period is specified in advance to the collector control unit **76** by, for example, an operator through the operation unit **62**.

However, the predetermined time period may be specified by the manufacturer of the machine tool **10**.

[0109] In addition, for example, according to the above embodiment, the acquisition unit **73** acquires the machining load LO based on the torque of the spindle motor **21**. However, the machining load LO may be calculated based on the drive current of the spindle motor **21**.

Therefore, the machine tool **10** may further include, for example, a current sensor for detecting the drive current of the spindle motor **21**. Thus, the acquisition unit **73** can acquire the information of the machining load LO based on the detection signal output from the current sensor.

[0110] Further, for example, according to the above embodiment, the relative movement speed VR is the relative movement speed VRX in the X direction or the relative movement speed VRY in the Y direction. However, the relative movement speed VR may be a composite speed of the relative movement speed VRX and the relative movement speed VRY.

[0111] Further, for example, the coolant discharge method is not limited to being that of the embodiment. For example, the coolant may be discharged using a center-through method. In that case, the coolant supply device **32** supplies the coolant to the spindle **18**. Further, the coolant may flow along the inner wall of the cover **30** (the machining area **48**).

[0112] For example, the machining device **12** may further be equipped with a non-illustrated recovery member in order to recover the coolant that falls downwardly of the table **26**. Such a recovery member, for example, is an oil pan provided on the pedestal **24**. A portion of the coolant supplied to the machining area **48** falls downwardly of the table **26** without turning into a mist.

According to the present exemplary modification, it is possible to collect the coolant that has fallen downwardly of the table **26**. The recovered coolant may be returned to the coolant tank **50**. In accordance with this feature, the coolant supply device **32** is capable of reusing the coolant that has been collected. In this instance, it is preferable for a filtering device (a filter) to be disposed between the recovery member and the coolant tank **50**. In accordance with this feature, a clean coolant can be returned to the coolant tank **50**.

[0113] At least one of the X-axis motor **47X** and the Y-axis motor **47Y** may be a linear motor. In the case where the X-axis motor **47X** is a linear motor, the encoder **29X** is a linear encoder. In the case where the Y-axis motor **47Y** is a linear motor, the encoder **29Y** is a linear encoder.

Invention That Can Be Understood from the Embodiments

[0114] The invention that can be grasped from the above embodiment and modifications are described below.

First Aspect of Invention

[0115] The first aspect of invention is the control device (**14**) for the machine tool (**10**) including the table (**26**) supporting the workpiece to be machined by the tool (**16**) in the machining area (**48**), the spindle (**18**) to which the tool is mounted and which rotates and moves relative to the table, and the mist collector (**34**) configured to collect mist in the machining area, the control device including the acquisition unit (**73**) configured to acquire at least one type of information from among the machining load (LO) applied to the tool by performing machining on the workpiece, the relative movement speed (VR) of the spindle with respect to the table, and the rotational speed (VS) of the spindle, the start determination unit (**75**) configured to determine whether or not to start the mist collector based on the at least one type of information acquired by the acquisition unit, and the collector control unit (**76**) configured to automatically control the mist collector based on the determination content of the start determination unit.

[0116] In accordance with such features, since the mist collector is automatically controlled depending on the content of machining, vibration of the mist collector is prevented from having a large adverse effect on the machine accuracy.

[0117] In the control device, the acquisition unit may acquire the information of the machining load based on the drive current or the torque of the spindle motor (**21**) of the spindle. In accordance with this feature, the mist collector can be turned ON/OFF automatically based on the machining load acquired during machining.

[0118] In the control device, the acquisition unit may acquire the information of the rotational speed based on the detection signal of the encoder (**25**) provided in the spindle motor (**21**) of the spindle. In accordance with this feature, the mist collector can be automatically turned ON and OFF in accordance with the rotational speed acquired during machining.

[0119] In the control device, the acquisition unit may acquire the information of the relative movement speed based on the detection signal of the encoder (**29**) provided in the feed axis motor (**47**) configured to relatively move the spindle and the table. In accordance with this feature, the mist collector can be automatically turned ON and OFF based on the relative movement speed obtained during machining.

[0120] The control device may further includes the estimation unit (**82**) configured to estimate the value, as the estimated value, of at least one of the machining load, the relative movement speed, and the rotational speed, based on the machining program (**72**) for performing the machining, wherein the acquisition unit acquires the estimated value as the information. Thus, it is possible to determine whether or not the mist collector is automatically started before the machining is started.

[0121] In the control device, the start determination unit may determine whether or not the predetermined start condition is satisfied, and determine not to start the mist collector in the case that the predetermined start condition is not satisfied, and the predetermined start condition may be satisfied in the case that at least one of the machining load, the relative movement speed, and the rotational speed acquired by the acquisition unit is equal to or greater than the threshold value

(TH). In accordance with such features, deterioration of the machine accuracy due to vibration of the mist collector is prevented.

[0122] In the control device, the threshold value may include the plurality of threshold values, and the start determination unit may use the threshold values different from each other in accordance with the machining load, the relative movement speed, and the rotational speed. In accordance with this feature, deterioration of the machine accuracy due to vibration of the mist collector is prevented.

[0123] In the control device, the start determination unit may determine to start the mist collector in the case that the predetermined start condition is satisfied. In accordance with this feature, the mist collector is automatically started, the mist is prevented from leaking out to the exterior of the machining area. Thus, in the case that there is little possibility that the machine accuracy is deteriorated due to the vibration of the mist collector, the mist can be collected by the mist collector.

[0124] The control device may further include the alarm output unit (**80**) configured to output the alarm in the case that an abnormality has occurred in the machine tool, and in the case that the alarm output unit has output the alarm, the collector control unit may prohibit the operation of the mist collector, regardless of the determination content of the start determination unit. In accordance with this feature, the mist collector is prevented from operating in the case that an abnormality has occurred in the machine tool.

Second Aspect of Invention

[0125] A second aspect of invention is a machine tool including the control device according to the first aspect of invention.

[0126] In accordance with this feature, since the mist collector is automatically controlled in accordance with the content of the machining, the vibration of the mist collector is prevented from having a large adverse effect on the machine accuracy.

[0127] The machine tool may further include the sub-control device (**78**) configured to control the mist collector instead of the collector control unit in the case that the control device is stopped. In accordance with this feature, even in the case that the control device is stopped, an automated control of the mist collector is carried out.

Third Aspect of Invention

[0128] A third aspect of invention is the control method for the machine tool (**10**) including the table (**26**) supporting the workpiece to be machined by the tool (**16**) in the machining area (**48**), the spindle (**18**) to which the tool is mounted and which rotates and moves relative to the table, and the mist collector (**34**) configured to collect mist in the machining area, the control method including the acquisition step (S1) in which the computer (**14**) acquires at least one type of information from among the machining load (LO) applied to the tool by performing machining on the workpiece, the relative movement speed (VR) of the spindle with respect to the table, and the rotational speed (VS) of the spindle, the start determination step (S2) in which the computer determines whether or not to start the mist collector based on the at least one type of information acquired in the acquisition step, and the collector control step (S3) in which the computer controls the mist collector based on the determination content in the start determination step.

[0129] In accordance with such features, since the mist collector is automatically controlled in accordance with the content of the machining, the vibration of the mist collector is prevented from having a large adverse effect on the machine accuracy.

REFERENCE SIGNS LIST

[0130] **10, 101**: machine tool [0131] **12**: machining device [0132] **14, 142, 143**: control device [0133] **16**: tool [0134] **18**: spindle [0135] **21**: spindle motor [0136] **25, 29**: encoder [0137] **26**: table [0138] **34**: mist collector [0139] **47**: feed axis motor [0140] **48**: machining area [0141] **72**: machining program [0142] **73**: acquisition unit [0143] **75**: start determination unit [0144] **76**: collector control unit [0145] **78**: sub-control device [0146] **80**: alarm output unit [0147] **82**:

estimation unit [0148] LO: machining load [0149] TH: threshold value [0150] VR: relative movement speed [0151] VS: rotational speed

Claims

1. A control device for a machine tool comprising a table supporting a workpiece to be machined by a tool in a machining area, a spindle to which the tool is mounted and which rotates and moves relative to the table, and a mist collector configured to collect mist in the machining area, the control device comprising: an acquisition unit configured to acquire at least one type of information from among a machining load applied to the tool by performing machining on the workpiece, a relative movement speed of the spindle with respect to the table, and a rotational speed of the spindle; a start determination unit configured to determine whether or not to start the mist collector based on the at least one type of information acquired by the acquisition unit; and a collector control unit configured to automatically control the mist collector based on a determination content of the start determination unit.
2. The control device according to claim 1, wherein the acquisition unit acquires the information of the machining load based on a drive current or a torque of a spindle motor of the spindle.
3. The control device according to claim 1, wherein the acquisition unit acquires the information of the rotational speed based on a detection signal of an encoder provided in a spindle motor of the spindle.
4. The control device according to claim 1, wherein the acquisition unit acquires the information of the relative movement speed based on a detection signal of an encoder provided in a feed axis motor configured to relatively move the spindle and the table.
5. The control device according to claim 1, further comprising an estimation unit configured to estimate a value, as an estimated value, of at least one of the machining load, the relative movement speed, and the rotational speed, based on a machining program for performing the machining, wherein the acquisition unit acquires the estimated value as the information.
6. The control device according to claim 1, wherein the start determination unit determines whether or not a predetermined start condition is satisfied, and determines not to start the mist collector in a case that the predetermined start condition is not satisfied, and the predetermined start condition is satisfied in a case that at least one of the machining load, the relative movement speed, and the rotational speed acquired by the acquisition unit is equal to or greater than a threshold value.
7. The control device according to claim 6, wherein the threshold value includes a plurality of the threshold values, and the start determination unit uses the threshold values different from each other in accordance with the machining load, the relative movement speed, and the rotational speed.
8. The control device according to claim 6, wherein the start determination unit determines to start the mist collector in a case that the predetermined start condition is satisfied.
9. The control device according to claim 1, further comprising an alarm output unit configured to output an alarm in a case that an abnormality has occurred in the machine tool, wherein, in a case that the alarm output unit has output the alarm, the collector control unit prohibits an operation of the mist collector, regardless of the determination content of the start determination unit.
10. A machine tool comprising the control device according to claim 1.
11. The machine tool according to claim 10, further comprising a sub-control device configured to control the mist collector instead of the collector control unit in a case that the control device is stopped.
12. A control method for a machine tool comprising a table supporting a workpiece to be machined by a tool in a machining area, a spindle to which the tool is mounted and which rotates and moves relative to the table, and a mist collector configured to collect mist in the machining area, the control method comprising: an acquisition step in which a computer acquires at least one type of

information from among a machining load applied to the tool by performing machining on the workpiece, a relative movement speed of the spindle with respect to the table, and a rotational speed of the spindle; a start determination step in which the computer determines whether or not to start the mist collector based on the at least one type of information acquired in the acquisition step; and a collector control step in which the computer controls the mist collector based on a determination content in the start determination step.
